



the government has launched some schemes like the solar pump scheme with a target of installing 100,000 solar pumps and programme to train 50,000 people for solar installations under the Surya Mitra scheme, no inter-state transmission charges and losses to be levied for solar and wind power, compulsory procurement of 100 per cent power from waste-to-energy plants, and renewable generation obligations on new thermal and lignite plants. Other significant initiatives include launch of improved cook-stoves; coordinated research and development activities in solar PV and thermal; and second-generation biofuels, hydrogen energy and fuel cells.

The ministry of New and Renewable Energy (MNRE) has taken some proactive steps to exemplify the government's dream in terms of green and clean energy. In this regard, the largest renewable energy capacity expansion plan in the world is being worked in India, with India now home to the world's largest solar plant in Kamuthi, Tamil Nadu. The Kamuthi plant has an electricity generation capacity of 648MW and covers an area of 10sq.m/km. It was built in eight months and is automated to the core, with robotic arms performing the cleaning duty daily. Prior to this, Topaz solar farm in California was the largest with 550MW capacity.

The ministry has earmarked certain key drivers for development of new and renewable energy platforms, which include energy security, electricity shortage, energy access, climate change and reduction in fossil fuel dependence. I think the steps taken in this direction are positive and sincere, and now with the industry already showing keen interest, the sector is slated to boom and take the new leap of faith.

Solving the energy security cauldron

Energy is a key ingredient to any developing economy, and for India it's one of the foremost. India has its

own challenges in terms of energy resources in proportion to its large population. While the country is home to nearly 17 per cent of the world's population, it has only 0.4, 0.4 and 6 per cent of the world's oil, gas and coal reserves, respectively. This has led to the dependence on a large import volume, which is nearly 35 per cent of the annual demand. To create a better value chain, efforts are on to explore and exploit India's oil, gas and coal reserves. However, there are innumerable structural issues.

India's renewable energy potential is vast and holds a great promise. It is evident that an integrated approach towards developing domestic energy resources, and giving special

attention to the ones in which India may have higher potential, has been engaging the attention of India's economic planners. On the other hand, there is a large scope for effecting efficiency in energy-demand sectors. Electrification of this demand could also ease India's import dependence, as there would be a shift from liquid fuels to coal and renewable energy. This is doable if India's renewable energy potential could be tapped, and a major demand shift could be effected in cooking and transport sectors, which are presently heavily dependent on petroleum products. There is also an immense possibility of electrifying irrigation pumps as well as telecom

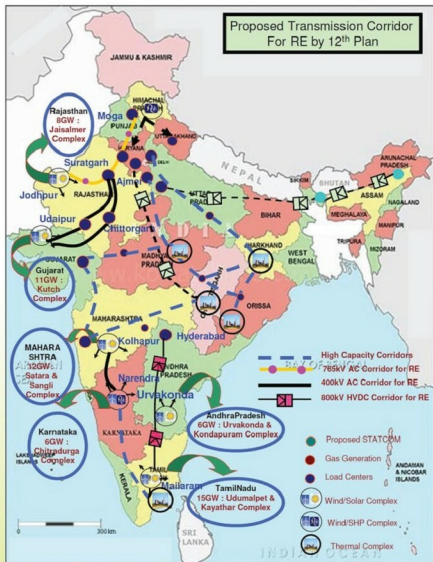


Fig. 2: Different energy corridors with plans